MIL-C-60805(MU)
15 October 1968
SUPERSEDING
FA-PD-MI-2346,Rev 0
4 May 1959

MILITARY SPECIFICATION
CARTRIDGE, IMPULSE: M150 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Impulse: M150 for use with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS

Munitions Command

D10521960	- Cartridge, Impulse, M150, Assembly
C8837888	- Carton, Packing, Ammunition for Cartridge, Impulse, M150

FSC 1377

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DRAWINGS (Cont'd)

C8837889 - Box, Packing, Ammunition for Cartridge, Impulse, M150
IEL-10521960 - Inspection Equipment List for Cartridge, Impulse, M150, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.4 Functional.

3.4.1 The cartridge shall not fail to fire.

3.4.2 Ballistics.- The cartridge when tested in an M8 Aircraft Canopy Remover shall comply with the ballistic requirements of Table I.

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Table I

Temperature Fahrenheit (F)	* Velocity min. (fps)	Peak Thrust Pounds Min.	Peak Thrust Pounds Max.	Ignition Delay (see 6.2)
-65°F to -70°F	10.7	3500	7000	25 milliseconds
65°F to 75°F	10.7	3500	7000	25 milliseconds
200°F to 205°F	10.7	3500	9000	25 milliseconds

* Velocity taken at tube separation.

3.4.2.1 Initiation.- The M8 Remover shall be fired by a source delivering an output equal to that of an M3A1 initiator at the end of 10 feet of aircraft hose, Size No. 4, in accordance with MS28741, at 70° ± 5°F.

3.5 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable

3.5.1 Metal defects.- The metal part shall be free from cracks, splits, laminations, perforations, porosity or other metal defects.

3.5.2 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

3.5.3 Burr.- The part or assembly shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contracr or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

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4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 units for hardness test from each heat treated lot, 23 units of each anodized lot for protective finish tests, 12 units for vibration test (from ballistic sample) and 21 ballistics samples and tested as indicated in Table II, and 10 sets of unassembled components (including explosive components) for visual and dimensional examination.

Table II

Paragraph Requirement	Test	Test	Sample Size	Type of Test Dest	Non-Dest
3.2	4.4.1	Protective finish	23	X	-
3.3	4.4.2	Vibration	12 (from ballistic sample)	-	X
Dwg.	4.4.3	Airtightness	Each	-	X
3.4.2	4.4.4	Ballistics	21	X	-
Dwg.	4.4.5	Hardness	50	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistics shall be temperature conditioned and comply with the requirements of Table I. The sample for firing shall consist of the following:

Table III

Temperature	Vibrated	Not Vibrated
-65° to -70°F	4	3
65° to 75°F	4	3
200° to 205°F	4	3

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4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submittal of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of 1200 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process
b. Propellant from not more than one lot
c. Igniters from not more than one lot
d. Primers from not more than one lot

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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Table V

Total Allowable Defects Per Sample Size

Sample Size for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Size for Minor Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Primer, Drawing D10551686, covering a detail of Drawing D10521960.

Defect	Method of Inspection
CRITICAL	
1. Improper assembly	Visual
2. Any part missing	Visual
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defects	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.2 Disc, Igniting, Drawing B8593355, covering a detail of Drawing D10521960.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage

4.3.2.3 Spring, Retaining, Drawing B8593629, covering a detail of Drawing D10521960.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter, outside, min.	Gage
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
201. Diameter, outside, max	Gage
202. Width of opening with spring compressed to specified opening	Gage
203. Thickness, max.	Gage

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4.3.2.4 Case, Cartridge, Drawing 10521961, covering a detail of Drawing D10521960.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of head, min.	Gage
104. Thickness of wall at head	Gage
105. Thickness of wall at base	Gage
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body	Gage
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Radius missing or incorrect	Visual
206. Surface finish improper	Visual

4.3.2.5 Head, Cartridge, Drawing C10521962, covering a detail of Drawing D10521960.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of forward shoulder (or body) with primer pocket	Gage
102. Concentricity of flange with primer pocket	Gage
103. Concentricity of large counterbore with primer pocket	Gage
104. Concentricity of body with primer pocket	Gage
105. Concentricity of undercut with body	Gage

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
106. Concentricity of inside diameter at mouth with body	Gage
107. Diameter of counterbore, max.	Gage
108. Diameter of flange, min.	Gage
109. Diameter of primer pocket	Gage
110. Diameter of flash hole	Gage
111. Depth of primer pocket	Gage
112. Depth from base of counterbore to base of flash hole	Gage
113. Protective finish inadequate or defective	Visual
114. Surface finish of primer pocket improper	Visual
115. Burr	Visual
116. Foreign matter	Visual
MINOR	
201. Diameter of counterbore, min.	Gage
202. Diameter at mouth of base	Gage
203. Diameter of flange, max.	Gage
204. Diameter of forward shoulder (or body)	Gage
205. Diameter of counterbore, min.	Gage
206. Diameter of main cavity	Gage
207. Diameter of body	Gage
208. Depth to base of counterbore	Gage
209. Depth of counterbore	Gage
210. Total length	Gage
211. Length of forward shoulder (or body)	Gage
212. Width of undercut	Gage
213. Width of counterbore	Gage
214. Thickness at base of flash hole	Gage
215. Location of datum diameter	Gage
216. Location of counterbore	Gage
217. Radius or chamfer missing or incorrect	Visual
218. Surface finish improper, except as otherwise classified	Visual

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4.3.2.6 Ring, Cartridge, Drawing B10521963, covering a detail of Drawing D10521960.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of inside diameter with body	Gage
102. Diameter of body, max.	Gage
103. Diameter, inside, min.	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Diameter of body, min.	Gage
202. Diameter, inside, max.	Gage
203. Thickness	Gage
204. Radius missing or incorrect	Visual
205. Surface finish improper	Visual

4.3.2.7 Gasket, Drawing B10521964, covering a detail of Drawing D10521960.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of inside diameter with body	Gage
102. Diameter of body, max.	Gage
103. Diameter, inside, min.	Gage
104. Material defective	Visual
105. Flash	Visual
106. Foreign matter	Visual

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of body, min.	Gage
202. Diameter inside	Gage
203. Thickness	Gage

4.3.2.8 Assembly, Cartridge, Impulse, M150 Assembly, Drawing D10521960.

CRITICAL	
1. Primer not properly seated	X-ray[1/] & Visual
2. Any part missing	X-ray[1/] & Visual
3. Improper assembly	X-ray[1/] & Visual
4. Metal defective	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Crimping and sealing of primer and primer pocket incorrect	Visual
MINOR	
201. Diameter of flange, max.	Gage
202. Diameter of body, max.	Gage
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Varnish missing	Visual

1/ The assembled cartridge shall be X-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.9 Carton, Packing, Ammunition for Cartridge, Impulse, M150, Drawing C8837888.

CRITICAL

None defined

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separators or fillers missing	Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

4.3.2.10 Box, Packing, Ammunition for Cartridge, Impulse, M150, Drawing C8837889.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of containers incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

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4.3.3.1 Protective finish.- Twenty three units from each anodized lot shall be tested. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Vibration.- Twelve cartridges from the initial production lot and twelve from each production lot (when there has been a change in the method of assembly or a different lot of components is used) shall be selected and tested. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.3 Airtightness.- Each cartridge shall be tested for pressure differential. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.4 Hardness.- Fifty units from each heat-treated lot shall be tested. Failure of any unit to comply with the requirements of the drawing shall be cause for rejection of the lot.

4.3.3.5 Functional.- The following tests shall be performed by the Government.

4.3.3.5.1 Ballistics.- Twenty-one assembled cartridges from each production lot shall be tested to determine compliance with requirement of 3.4.1 and 3.4.2. The units shall be temperature conditioned as provided in Table VI and 4.4.4.1. The variables sampling plan of MIL-STD-414, Example C-2 or C-3, as applicable for the AQL listed in Table VI shall be used for each sample of 7 units. Failure of any cartridge to comply with the requirements of 3.4.1 and 3.4.2 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10521960.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and cartridge head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.2 Vibration.- Test procedures for the vibration test shall be in accordance with MIL-E-5272.

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4.4.3 Airtightness.- The method for determining airtightness shall be by use of equipment listed on IEL-10521960.

4.4.4 Ballistics.- The test sample shall be rigidly supported in the test fixture and shall be grouped, temperature conditioned and tested. Grouping and temperature conditioning shall be in accordance with Table VI and 4.4.4.1.

Table VI

Temperature (°)(F)	AQL	Sample Size
65 to 75	0.65%	7
-65 to -70	0.65%	7
200 to 205	0.65%	7

4.4.4.1 Temperature conditioning.- Cartridges shall be loaded into acceptable M8 Canopy Removers (or reasonably like facsimile). Caps shall be inserted on all gas port openings during the conditioning period. The temperature of the propellant within the cartridge shall be considered as the conditioning temperature and will be determined by means of a thermocupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the removers being used as test fixtures. The conditioning time shall be one hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature. The sample shall be tested within 10 minutes after removal from the conditioning chamber. The sample shall be reconditioned in accordance with 4.4.4 and 4.4.4.1 if this time is exceeded. The total conditioning time shall not exceed 18 hours.

4.4.5 Hardness.- A hardness testing machine shall be used to determine the values specified on the applicable drawing.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8837888 and C8837889.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

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6.2 Ignition delay is defined as the time that elapses between firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2346, Rev. 0 dated 4 May 1959.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A083

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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